PHYS 4D Exam 2 Reference Sheet (with LaTeX)

Write Units

Constants

$$h = 6.626 \times 10^{-34}\,\mathrm{J\cdot s} \qquad \hbar = 1.054 \times 10^{-34}\,\mathrm{J\cdot s}$$

$$c = 2.998\,\mathrm{m/s} \qquad e^{\pm} = \pm 1.602 \times 10^{-19}\,\mathrm{C}$$

$$m_e = 9.11 \times 10^{-31}\,\mathrm{kg} \qquad m_e = 0.511\,\mathrm{MeV/c^2}$$

$$\varepsilon_0 = 8.854 \times 10^{-12}\,\frac{\mathrm{C^2}}{\mathrm{Nm^2}} \qquad \mu_0 = 4\pi \times 10^{-7}\,\mathrm{N/A^2}$$

General Waves

$$\left(\nabla^2 \vec{\psi} =\right) \frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 \psi}{\partial t^2} v = \lambda f = \frac{\omega}{k}$$

$$\psi(x,t) = \psi_m \cos(kx \mp \omega t); f = \frac{\omega}{2\pi} = \frac{1}{T}; k = \frac{2\pi}{\lambda}$$

Consine can be exchanged with another function like $\sin(\theta)$ or $e^{i\theta}$.
Superposition of waves also fulfills wave equation.
Pulses tend to follow the Gaussian
$(e^{-\alpha t}; \int_{-\infty}^{\infty} e^{-\alpha t}\,dt = \sqrt{\frac{\pi}{\alpha}})$

Standing waves

Two traveling waves in opposite directions

$$\psi(x,t) = \psi_m \cos(kx + \phi)\cos(\omega t + \phi') = \psi_m \left(e^{i(kx-\omega t)} + e^{i(kx+\omega t)}\right)$$

Photoelectric Effect Photons hitting electrons beyond a certain energy can cause electrons to break free

$$E=hf=\hbar\omega; KE_{max}=hf-\phi=eV_{stop}; I=\frac{1}{2\mu_0 c}\vec{E}_0^2$$

Bohr Model Proposed quantization of electron angular momentum (1 QN: $rmv = n\hbar$ for $n \in \mathbb{N}$)
Explains emission spectrum of atoms
Derived radius & energy formulas from velocity

$$r_n = \frac{4\pi\varepsilon_0\hbar}{me^2}n^2; E_n = -\frac{me^4}{32\pi^2\varepsilon_0^2\hbar^2}\frac{1}{n^2} \approx \frac{-13.6\mathrm{eV}}{n^2}$$

Bohr radius a_0 is r_n when $n = 0$

Compton Effect Shoot photon (λ) at electron
Emits photon (λ') at angle θ

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta); \frac{h}{m_e c} = 0.002426\,\mathrm{nm}$$

De Broglie Wavelength: Quantum particles fired at a slit behave like a wave of wavelength $\lambda = \frac{h}{p}$

Schrodinger's 1D Equation: Allows 3 QN

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(x)\right]\psi(x) = E\psi(x)$$

$\vec{F} = -\nabla V$; ψ is probability wave; also used to get averages (function or operator f) or constants

$$P = 1 = \int_{-\infty}^{\infty} \psi^\star \psi\,dx; \langle f\rangle = \int_{-\infty}^{\infty} \psi^\star f \psi\,dx$$

Infinte Well

$$V(x) = \begin{cases} 0 & \text{if } a < x < b \\ \infty & \text{otherwise}\end{cases}; \psi(x) = Ae^{ik}$$

$$k_n^2 = \frac{n^2\pi^2}{(b-a)^2} = \frac{2mE_n}{\hbar^2}; E_n = \frac{n^2\pi^2\hbar^2}{2m(b-a)^2}$$

Finite Well Transcendental algebra problem

$$V(x) = \begin{cases} 0 & \text{if } a < x < b \\ V_0 & \text{otherwise}\end{cases}$$

Waves either reflect or transmit at edge

$$T = \frac{4k_1k_2}{(k_1+k_2)^2}; R = \frac{(k_1-k_2)^2}{(k_1+k_2)^2}; T + R = 1$$

Harmonic Oscillator $V(x)$ follows Taylor series or
$V(x) = \frac{1}{2}kx^2$; uses polynomial H_n

$$\psi_n(x) = A_n H_n e^{-ax^2}; a = \frac{\sqrt{km}}{2\hbar}; E_n = \left(n + \frac{1}{2}\right)\hbar\sqrt{\frac{k}{m}}$$

Operators Basically eigenvectors and eigenvalues just with functions and not vectors.

$$\hat{A}\psi = A\psi; \hat{p} = -i\hbar\nabla; \hat{E} = -\frac{\hbar^2}{2m}\nabla^2 + V(x)$$

Heisenberg Uncertainty Principle

$$\Delta x\Delta p \geq \hbar/2; \Delta t\Delta E \geq \hbar/2$$

Fourier Analysis: An extra thing you can use for Laplace transforms

$$\psi(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} g(k)e^{ikx}\,dk; g(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\psi(k)e^{-ikx}\,dk$$

Hydrogen Atom: 3D; Uses spherical coordinates

$$x = r\sin\theta\cos\phi; y = r\sin\theta\sin\phi; z = r\cos\theta$$

$$r = \sqrt{x^2+y^2+z^2}; \theta = \arccos\frac{z}{r}; \phi = \arctan\frac{y}{x}$$

Schrodinger's can be turned spherical.

$$\left[\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin\theta}\frac{\partial^2}{\partial\theta^2}\right]\psi = E\psi$$

Solution becomes seperable; $\Theta(\theta)$ & $\Phi(\phi)$ in exam

$$\psi(r,\theta,\phi) = R(r)\cdot\Theta(\theta)\cdot\Phi(\phi)$$

Name	Used in	Values	Transition
n	$R(r)$	$\mathbb{N}$	N/A
ℓ	$R(r)$, $\Theta(\theta)$, $\vec{L}$	$\mathbb{N} < n$	$\Delta\ell = \pm 1$
m_ℓ	$\Theta(\theta)$, $\Phi(\phi)$	$\mathbb{N}; -\ell \leq m_\ell \leq \ell$	$\Delta m_\ell \in -1, 0, 1$

Quantum Angular Momentum ($\vec{L}$)

$$L = \hbar\sqrt{\ell(\ell+1)}; L_z = m_\ell\hbar$$

Many-Electron Atoms

$$E_n = (-13.6\,\mathrm{eV})\frac{Z_{eff}^2}{n^2}; Z_{eff} = q_{nucleus} - q_{screeners}$$

For transition energies:

$$\Delta E = 13.6\,\mathrm{eV}(Z-1)^2\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

For electrons in incomplete shells, exists angular (L) & magnetic (M_L) Q.N's:

$$|L_i| = \hbar\sqrt{l_i(l_i+1)}; L_{max} = l_1 + l_2; L_{min} = |l_1 - l_2|$$

$$L \in [L_{min}, L_{max}] \cap \mathbb{N}; M_L = \sum m_{l,i}; M_L \in [-L, L] \cap \mathbb{N}$$

S can be computed same as M_L but with spin quantum number
Hund's rule: $L = M_{L,max}$, $S = M_{S,max}$; maximize total spin
$\Delta L \in \{0, \pm 1\}; \Delta S = 0$
HeNe laser produces photons

$$\lambda = 632.8\,\mathrm{nm}; E = 1.96\,\mathrm{eV}; E_0 = \sqrt{\frac{2I}{c\varepsilon_0}}; I = \frac{P}{A}$$

Molecular Structure

$$E(H + H^+) = E(H) + E(H^+) = -13.6\,\mathrm{eV} + 0; E(H_2^+) = -16.3\,\mathrm{eV}; E(H_2) = -31.7\,\mathrm{eV}$$

Covalent bonded atoms share electrons, forming bonding then antibonding
Bond dissociation energy inversely proportional to periodic y-coord and equilibrium separation, proportional to periodic x-coord
Ionic bonds come from 2 ions pulled by Coulombic force
Pauli exclusion prevents 2 atoms from being too close; has potential energy

$$E_{molecule} = U_R + U_C + \Delta E_{ionization}; U_C = -\frac{q_1q_2}{4\pi\varepsilon_0 R_{eq}}$$

Can be considered electric dipole of moment p, often off from measured value

$$p = qr = qR_{eq}$$

Vibrational states in molecules:

$$E_N = (N + \frac{1}{2})\hbar\omega; \omega = \sqrt{\frac{k}{m}} = 2\pi f; \Delta N = \pm 1$$

Use harmonic mean for mass

$$m = \left(\sum \frac{1}{m_i}\right)^{-1} \left[= \frac{m_1 \cdot m_2}{m_1 + m_2}\right]$$

Rotational energy also exists, from Q.N. L

$$E_L = \frac{L(L+1)\hbar^2}{2mR_{eq}^2}; \Delta L = \pm 1$$

Statistical Physics
Indistinguishable particles measure probability of finding one with specific energy E; Uses microstate weight W_i; $U(E)$ is energy density at energy E

$$p(E) = \frac{\sum_i N_{E,i} W_i}{N_{total} \sum_i W_i}$$
$$U(E) = Eg(E)f(E); dN = N(E)\,dE = Vg(E)f(E)\,dE$$

Can find density of particles in energy state E

$$g_{mass}(E) = \frac{4\pi(2s+1)\sqrt{2}(mc^2)^{3/2}}{(hc)^3}; g_\gamma(E) = \frac{E^2}{\pi^2(\hbar c)^3} = \frac{8\pi E^2}{(hc)^3}$$

For dilute gas, go straight to Maxwell-Boltzmann.

$$k = 1.38 \times 10^{-23}\text{ J/K} = 8.617 \times 10^{-5}\text{ eV/K}; K_{avg} = \frac{3}{2}kT$$
$$N(E) = \frac{2N}{\sqrt{\pi}(kT)^{3/2}} E^{\frac{1}{2}} e^{-\frac{E}{kT}}; N(v) = N\sqrt{\frac{2}{\pi}} \left(\frac{m}{kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}}$$

Approximate tiny distribution integrals.

$$\int_{E_m}^{E_m+\Delta E} N(E)dE \approx N(E) * \Delta E$$

Doppler Broadening of Spectral Line

$$\Delta f = 2f_0\sqrt{(2\ln(2))kT/mc^2}$$

Blackbody radiation energy distribution comes from Bose-Einstein distribution

$$f_{BE}(E) = \frac{1}{A_{BE}e^{E/kT} - 1}; U = \frac{8\pi^5 k_B^4}{15(hc)^3}T^4$$

Fermi-Dirac energy uses Fermi energy E_F

$$f_{FD}(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}; E_F = \frac{h^2}{2m}\left(\frac{3N}{8\pi V}\right)^{\frac{2}{3}}; E_m = \frac{3}{5}E_F$$

Relativity Δt_0 is proper time; L_0 = proper length

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}; L = L_0\sqrt{1 - u^2/c^2}$$
$$v = \frac{v' - u}{1 + v'u/c^2}; f' = f\sqrt{\frac{1 - u/c}{1 + u/c}}$$

Lorentz Transforms

$$x' = \frac{x - ut}{\sqrt{1 - u^2/c^2}}; \begin{matrix} y' = y; \\ z' = z; \end{matrix} t' = \frac{t - (u/c^2)x}{\sqrt{1 - u^2/c^2}}$$
$$v_x' = \frac{v_x - u}{1 - v_x u/c^2}; v_y' = \frac{v_y\sqrt{1 - u^2/c^2}}{1 - v_x u/c^2} v_z' = \frac{v_z\sqrt{1 - u^2/c^2}}{1 - v_x u/c^2}$$

Relativity alters momentum and energy; total remains constant

$$\vec{p} = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}}; E = K + E_0 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}; E_0 = mc^2$$
$$E^2 = (pc)^2 + (mc^2)^2; E \approxeq pc$$